

Project title: Dynamic investment decisions in 6G infrastructure under carbon emission constraints: a real-options perspective

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Context:

The telecommunications industry has a 10-years cycle with a new generation of mobile networks released roughly every decade. While 5G deployment started few years ago and 5G networks did not yet reach maturity in many countries, standardization of 6G technologies is ongoing in 3GPP, allowing for new spectrum bands (more capacity) and new physical layer techniques (higher spectrum and energy efficiency) [1].

This technological race raises serious sustainability concerns. First, investing in new infrastructure under the principle of “If you build it, they will come”, is no more economically sustainable, due to the maturity of the retail telecommunications market in developed countries and the reluctance of vertical industries in deploying specialized, and costly, 5G infrastructure. Second, environmental sustainability is a new important constraint for the telecommunications industry, with a net-zero commitment by 2040-2050 for most operators. Economical, societal and environmental sustainability should indeed be the main driver for 6G development, rather than pure technological advances [2].

Objective:

The objective of this project is to develop a dynamic decision-making framework for 6G network investment, incorporating uncertainty related to revenues and regulations. Revenues are highly uncertain in mobile networks due to the future deployment of data-intensive AI services. Market dynamics add a second level of uncertainty, as the integration of mobile edge cloud infrastructure is a potential lucrative perspective, but is subject to deployment of advanced AI services by vertical industries. Finally, a third level of uncertainty is due to regulations, as they may transform the voluntary net-zero commitments into hard carbon emission targets.

The project will model investment decisions in mobile infrastructure under the lens of **real options**, as they are subject to large uncertainties and are largely irreversible. The model in [3] may be extended to integrate the new sources of uncertainty, related to regulations and market dynamics. A dynamic programming solution for the modeled real option is to be proposed, either considering an objective function that incorporates a sustainability metric, or under a hard carbon emission constraint.

References:

- [1] Chen, Wanshi, et al. "Toward standardization of 6G and NextG: Key technologies to enable fundamental enhancements." *IEEE Journal on Selected Areas in Communications* 44 (2026): 4333-4365.
- [2] Elayoubi, Salah Eddine, and Eric Hardouin. "For an energy sober network that contributes to a sustainable society." 2023 IEEE Future Networks World Forum (FNWF). IEEE, 2023.
- [3] Morlot, Frédéric, Salah Eddine Elayoubi, and Thomas Redon. "Risk-hedging using options for upgrading investments in mobile networks." 6th International ICST Conference on Performance Evaluation Methodologies and Tools. IEEE, 2012.